# Supplementary Figures

**Fig. S1** | *tbd* Map of 35 islands surveyed by CREP across the tropical Pacific.

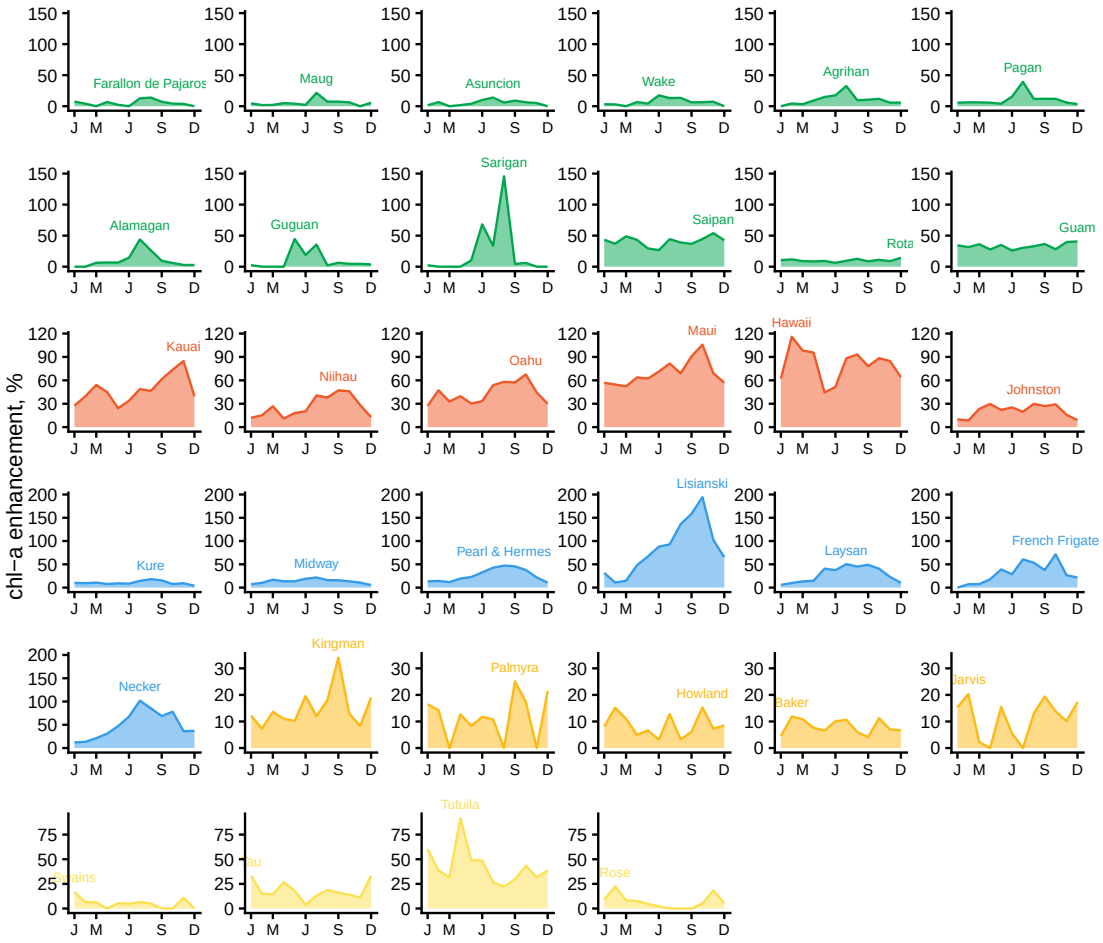


**Fig. S2** | Seasonal variation in IME strength at each island complex. Lines are mean-monthly chl-a enhancement from January to December, ordered and coloured by island group.

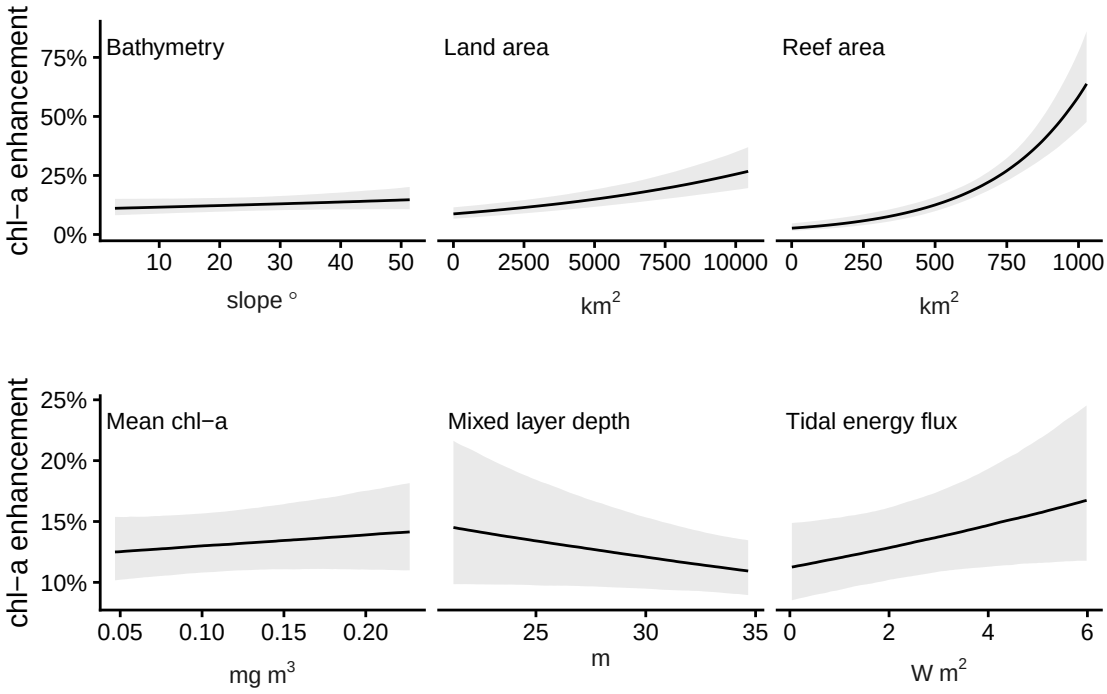


**Fig. S3** | Spatial effects on chl-a enhancement for geomorphic features (top) and average oceanographic conditions (bottom) Lines are median posterior predictions (± 95% certainty intervals). Note different y-axis values between static and dynamic covariates.

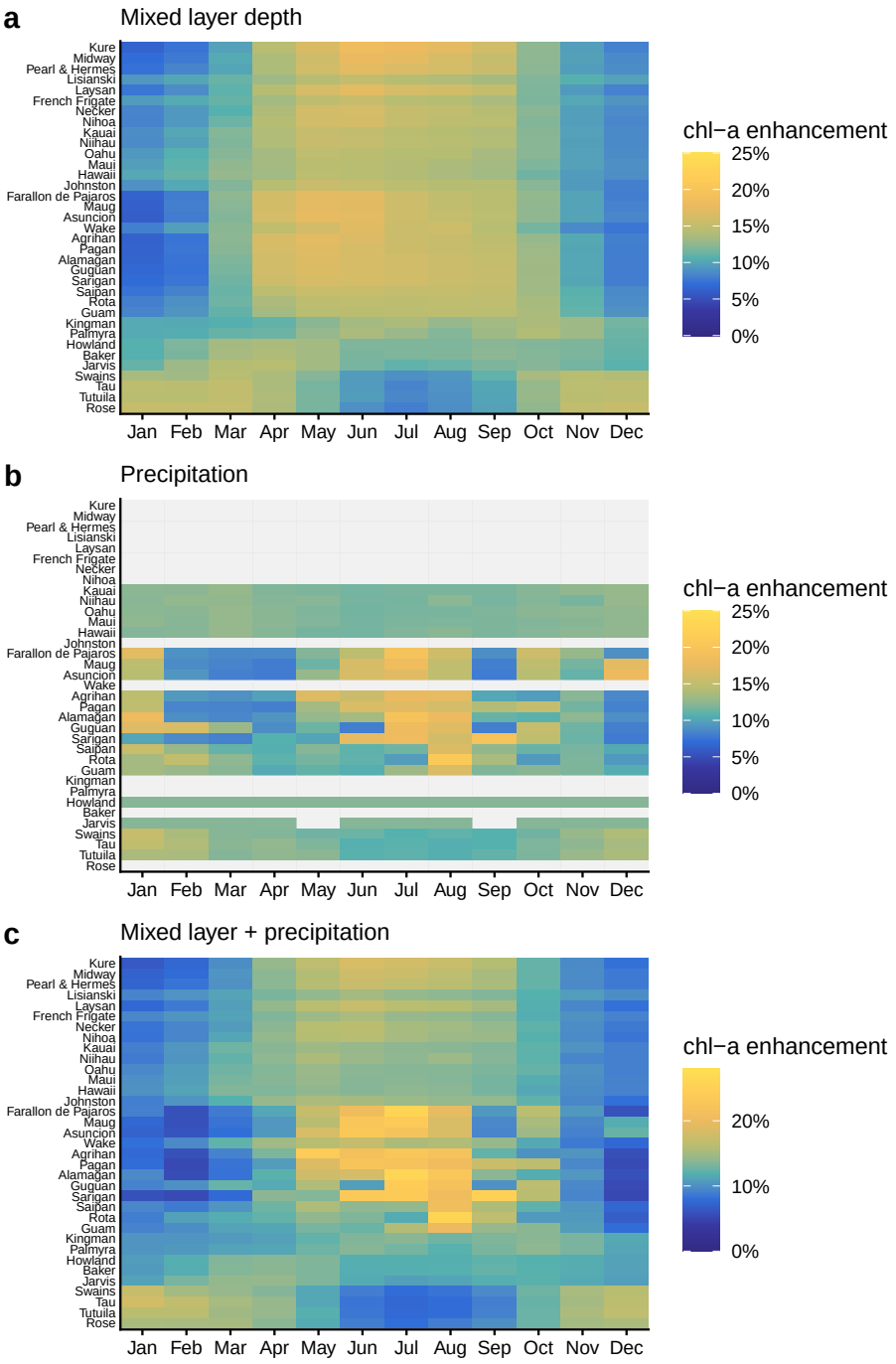


**Fig. S4** | Heatmap of seasonal effects on chl-a enhancement for a mixed layer depth, b precipitation and c both covariates combined. Islands with less than 1 mm monthly rainfall are shown in grey. All values are median posterior predictions excluding effects of other covariates.

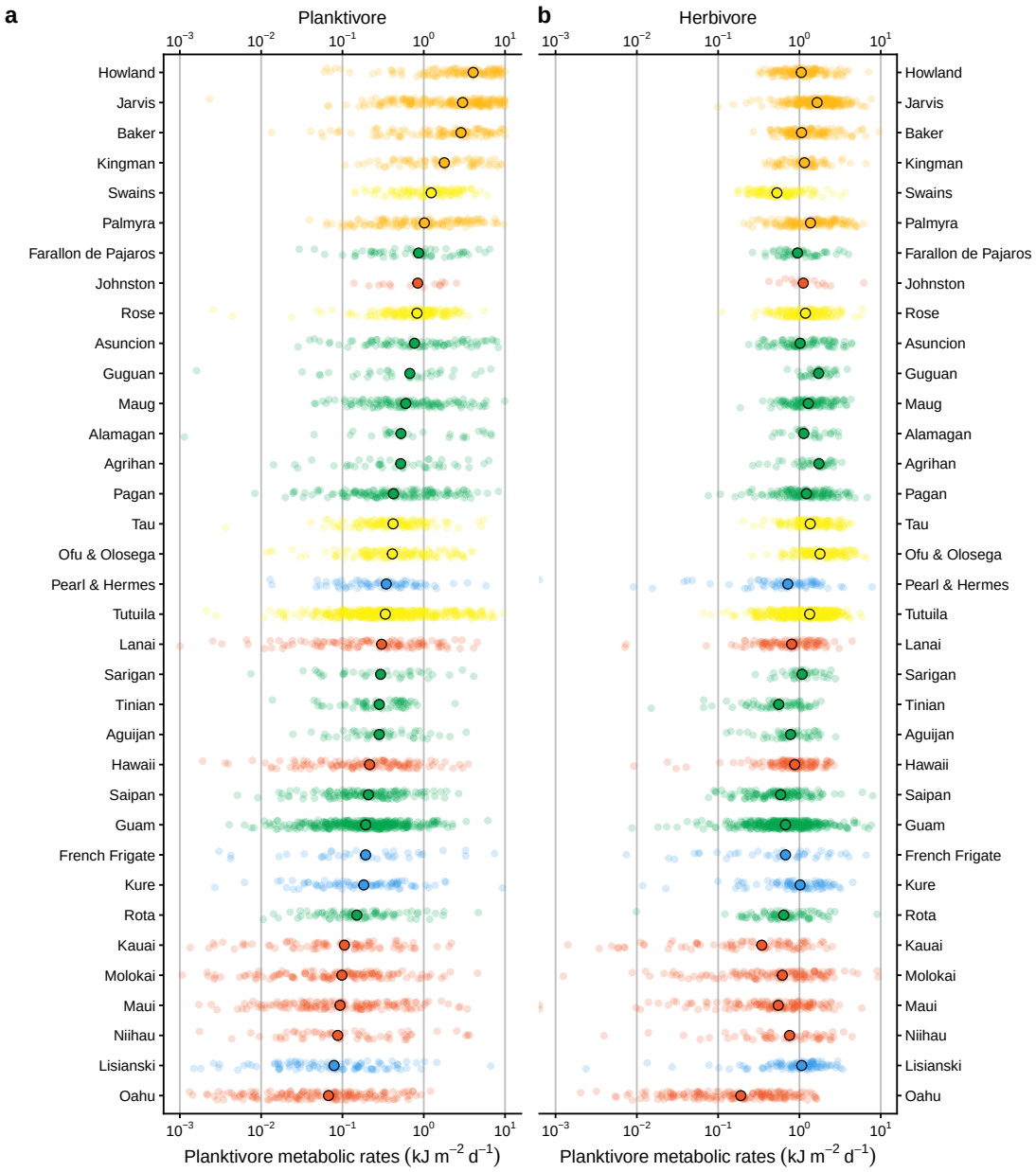


**Fig. S5** | Metabolic rate of reef-associated planktivores on coral reefs at 39 Pacific islands and atolls. Each point is a site-year combination (n = XX).